STAT332: Assignment 2 Question 5

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5. Using the survey package with weight variable finalwt and properly accounting that SYC used a stratified sampling design, give a point estimate and corresponding 95% CI for each of the following:

```
library("survey")
syc <- read.table("syc.txt", header = TRUE, sep = ",")

# Handle missing codes
syc$age[syc$age == 99] <- NA
syc$race[syc$race == 9] <- NA
syc$numarr[syc$numarr == 99] <- NA
syc$prviol[syc$prviol == 9] <- NA
syc$everdrug[syc$everdrug == 9] <- NA

# Survey design object
syc_des <- svydesign(
    ids = ~1,
    strata = ~stratum,
    weights = ~finalwt,
    data = syc
)
df0 <- degf(syc_des)
```

(a) Estimate the average age of juveniles/young adults in custody.

```
a <- svymean(~age, syc_des, na.rm = TRUE)
confint(a, df = df0)
```

```
##        2.5 %   97.5 %
## age 16.57537 16.70322
```

```
a
```

```
##       mean     SE
## age 16.639 0.0326
```

The point estimate for the average age of all juveniles/young adults in custody is 16.639 with a 95% confidence interval [16.5754, 16.7032].

(b) Estimate the mean number of prior arrests.

```r
b <- svymean(~numarr, syc_des, na.rm = TRUE)
confint(b, df = df0)
```

```
##           2.5 %   97.5 %
## numarr 8.418105 9.441383
```

```r
b
```

```
##          mean     SE
## numarr 8.9297 0.2609
```

The point estimate for the mean number of prior arrests is 8.930 with a 95% confidence interval $[8.4181, 9.4414]$.

(c) Estimate the proportion of juveniles/young adults in custody that have used illegal drugs.

```r
c <- svymean(~as.factor(everdrug), syc_des, na.rm =TRUE)
confint(c, df = df0)
```

```
##                           2.5 %    97.5 %
## as.factor(everdrug)0 0.1546300 0.1889484
## as.factor(everdrug)1 0.8110516 0.8453700
```

```r
c
```

```
##                          mean     SE
## as.factor(everdrug)0 0.17179 0.0088
## as.factor(everdrug)1 0.82821 0.0088
```

The point estimate for the proportion of juveniles/young adults in custody that have used illegal drugs is 82.821% with a 95% confidence interval $[81.1052, 84.5370]$.\

(d) Estimate the proportion of juveniles/young adults in custody that were previously arrested for a violent crime

```r
d <- svymean(~prviol, syc_des, na.rm = TRUE)
confint(d, df = df0)
```

```
##            2.5 %    97.5 %
## prviol 0.3338345 0.3753451
```

```r
d
```

```
##           mean     SE
## prviol 0.35459 0.0106
```

The estimated proportion of juveniles/young adults in custody that were previously arrested for a violent crime is 35.459%. The 95% confidence interval for the proportion is $[33.3835\%, 37.5345\%]$.

(e) Estimate the proportion of juveniles/young adults in custody that have used illegal drugs and were previously arrested for a violent crime.

```r
# create joint indicator variable
syc$vio_and_drug <- ifelse(syc$everdrug == 1 & syc$prviol == 1, 1, 0)
syc_des <- svydesign( # redefine design object
    ids = ~1,
    strata = ~stratum,
    weights = ~finalwt,
    data = syc
)
e <- svymean(~vio_and_drug, syc_des, na.rm = TRUE)
confint(e, df = df0)
```

```
##                  2.5 %    97.5 %
## vio_and_drug 0.2939274 0.3338513
```

```r
e
```

```
##                 mean     SE
## vio_and_drug 0.31389 0.0102
```

The estimated proportion of juveniles/young adults in custody that have used illegal drugs and were previously arrested for a violent crime is 31.389% with 95% confidence interval $[29.3927, 33.3851]$.\

(f) Estimate the proportion of males amongst juveniles/young adults in custody.

```r
f <- svymean(~factor(sex), syc_des, na.rm = TRUE)
confint(f, df = df0)
```

```
##                   2.5 %     97.5 %
## factor(sex)1 0.92051019 0.94187586
## factor(sex)2 0.05812414 0.07948981
```

```r
f
```

```
##                  mean     SE
## factor(sex)1 0.931193 0.0054
## factor(sex)2 0.068807 0.0054
```

The estimated proportion of males amongst juveniles/young adults in custody is 93.119% with 95% confidence interval $[92.0510, 94.1876]$.\

(g) Estimate the proportion of African Americans amongst juveniles/young adults in custody.

```r
g <- svymean(~factor(race), syc_des, na.rm = TRUE)
confint(g, df = df0)
```

```
##                      2.5 %     97.5 %
## factor(race)1 0.505688065 0.54965629
## factor(race)2 0.393550305 0.43705516
## factor(race)3 0.004913586 0.01205388
## factor(race)4 0.019799330 0.03318398
## factor(race)5 0.015723322 0.02837608
```

```r
g
```

```
##                      mean     SE
## factor(race)1 0.5276722 0.0112
## factor(race)2 0.4153027 0.0111
## factor(race)3 0.0084837 0.0018
## factor(race)4 0.0264917 0.0034
## factor(race)5 0.0220497 0.0032
```

The estimated proportion of African Americans amongst juveniles/young adults in custody is 41.5303% with 95% confidence interval $[39.3550, 43.7055]$.\

(h) Estimate the proportion of African American males amongst juveniles/young adults in custody.

```r
syc$black_male <- ifelse(syc$race == 2 & syc$sex == 1, 1, 0)
syc_des <- svydesign( # redefine design object
    ids = ~1,
    strata = ~stratum,
    weights = ~finalwt,
    data = syc
)
h <- svymean(~black_male, syc_des, na.rm = TRUE)
confint(h, df = df0)
```

```
##                2.5 %   97.5 %
## black_male 0.3760579 0.419252
```

```r
h
```

```
##               mean    SE
## black_male 0.39765 0.011
```

The estimated proportion of African American males amongst juveniles/young adults in custody is 39.765% with 95% confidence interval $[37.6058, 41.9252]$.

(i) From census data, we know that African-American males are about 7.6% of the youth population in the U.S. Using your answer to part (h), what can you conclude about racial bias when it comes to youth in custody?

African-American males are overrepresented in youth custody by roughly 5 times their share of the general youth population. This suggests substantial racial disparities and perhaps systemic bias in the youth justice system.